

Suggested Materials & Methods text for describing Olink Target 96/48 panel in publications

Original:

Suggested Materials & Methods text for describing Olink in publications

Proteins were measured using the Olink® XXXXXXXX panel* (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. The Proximity Extension Assay (PEA) technology used for the Olink protocol has been well described (Assarsson et al, 2014), and enables 92 analytes to be analyzed simultaneously, using 1 µL of each sample. In brief, pairs of oligonucleotide-labeled antibody probes bind to their targeted protein, and if the two probes are brought in close proximity the oligonucleotides will hybridize in a pair-wise manner. The addition of a DNA polymerase leads to a proximity-dependent DNA polymerization event, generating a unique PCR target sequence. The resulting DNA sequence is subsequently detected and quantified using a microfluidic real-time PCR instrument (Signature Q100). Data is then quality controlled and normalized using an internal extension control and an inter-plate control, to adjust for intra- and inter-run variation. The final assay read-out is presented in Normalized Protein eXpression (NPX) values, which is an arbitrary unit on a log₂-scale where a high value corresponds to a higher protein expression. All assay validation data (detection limits, intra- and inter-assay precision data, etc.) are available on manufacturer's website (www.olink.com).

* Previously branded as Olink® Multiplex panels

Long:

Suggested Materials & Methods text for describing Olink Proteomics in publications

Proteins were measured using the Olink® XXXXXXXX panel* (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. The Proximity Extension Assay (PEA) technology used for the Olink protocol has been well described (Assarsson et al, 2014), and enables 92 analytes to be analyzed simultaneously, using 1 µL of each sample. In brief, pairs of oligonucleotide-labeled antibody probes bind to their targeted protein, and if the two probes are brought in close proximity the oligonucleotides will hybridize in a pair-wise manner. The addition of a DNA polymerase leads to a proximity-dependent DNA polymerization event, generating a unique PCR target sequence. The resulting DNA sequence is subsequently detected and quantified using a microfluidic real-time PCR instrument (Signature Q100). The resulting Ct-data is then quality controlled and normalized using a set of internal and external controls. The final

assay read-out is presented in Normalized Protein eXpression (NPX) values, which is an arbitrary unit on a log₂-scale where a high value corresponds to a higher protein expression. The four internal controls are designed to mimic and monitor the different steps of the PEA. They consist of two non-human proteins with matching antibody-probes that functions as incubation/immuno controls, an IgG antibody with two matching probes attached to it that functions as an extension control and a complete double-stranded amplicon that function as detection control. The internal controls are introduced to all samples as well as to the external controls. The external controls consist of a triplicate of negative controls used to calculate the limit of detection (LOD), as well as a triplicate of interplate controls (IPCs) containing 92 sets of antibodies with both matching probes for each assay attached to them that are used for normalization. For each sample and assay NPX is calculated by the following equations: 1. $Ct(\text{analyte}) - Ct(\text{extension control}) = dCt(\text{analyte})$ (to decrease technical variation) 2. $dCt(\text{analyte}) - Ct(\text{median IPC}) = ddCt(\text{analyte})$ (to improve inter plate variation) 3. $\text{Correction factor}(\text{analyte}) - ddCt(\text{analyte}) = \text{NPX}(\text{analyte})$ (for more intuitive data). The correction factor is a set variable unique for each assay and reagent lot.

Quality control of the data is performed in two steps: First, the standard deviation for each of the incubation/immuno controls and the detection control is calculated for each run. A run will only pass quality control if the standard deviation for each of the controls is below 0.2. Secondly, each sample is quality controlled using incubation control 2 and the detection control. The run median of each of the controls is calculated and all samples within the run is compared to that. Samples that fall more than +/- 0.3 NPX from the plate median with regards to these two controls will fail the quality control and receive a QC warning in the data output file.

All assay validation data (detection limits, intra- and inter-assay precision data, etc.) are available on manufacturer's website (www.olink.com).

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Short:

Suggested Materials & Methods text for describing Olink in publications

Proteins were measured using the Olink® XXXXXXXX panel* (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. The Proximity Extension Assay (PEA) technology used for the Olink protocol has been well described (Assarsson et al, 2014), and enables 92 analytes to be analyzed simultaneously, using 1 µL of each sample. In brief, pairs of oligonucleotide-labeled antibody probes bind to their targeted protein, and if the two probes are brought in close proximity the oligonucleotides will hybridize in a pair-wise manner. The addition of a DNA polymerase leads to a proximity-dependent DNA polymerization event, generating a unique PCR target sequence. The resulting DNA sequence is subsequently detected and quantified using a microfluidic real-time PCR instrument (Signature Q100,). The resulting Ct-data is then quality controlled and normalized using a set of internal and external controls. The final

assay read-out is presented in Normalized Protein eXpression (NPX) values, which is an arbitrary unit on a log2-scale where a high value corresponds to a higher protein expression.

The internal controls are designed to mimic and monitor the different steps of the PEA. They consist of two incubation/immuno controls, an extension control and a detection control . The internal controls are introduced to all samples as well as to the external controls and are used for quality control and normalization of the data. The external controls consist of a negative control used to calculate the limit of detection (LOD), as well as a triplicate of interplate controls (IPCs) that are used for data normalization. Quality control of the data is performed in two steps: First, the run is quality controlled by calculating the standard deviation for the detection control and the incubation/immuno controls. The standard deviation should be below 0.2 for a run to pass quality control. Secondly, each sample is quality controlled by comparing the results for the detection control and one of the incubation controls against the run median. Samples that fall more than 0.3 NPX from the run median with regards to these two internal controls will fail the quality control.

All assay validation data (detection limits, intra- and inter-assay precision data, etc.) are available on manufacturer's website (www.olink.com).

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